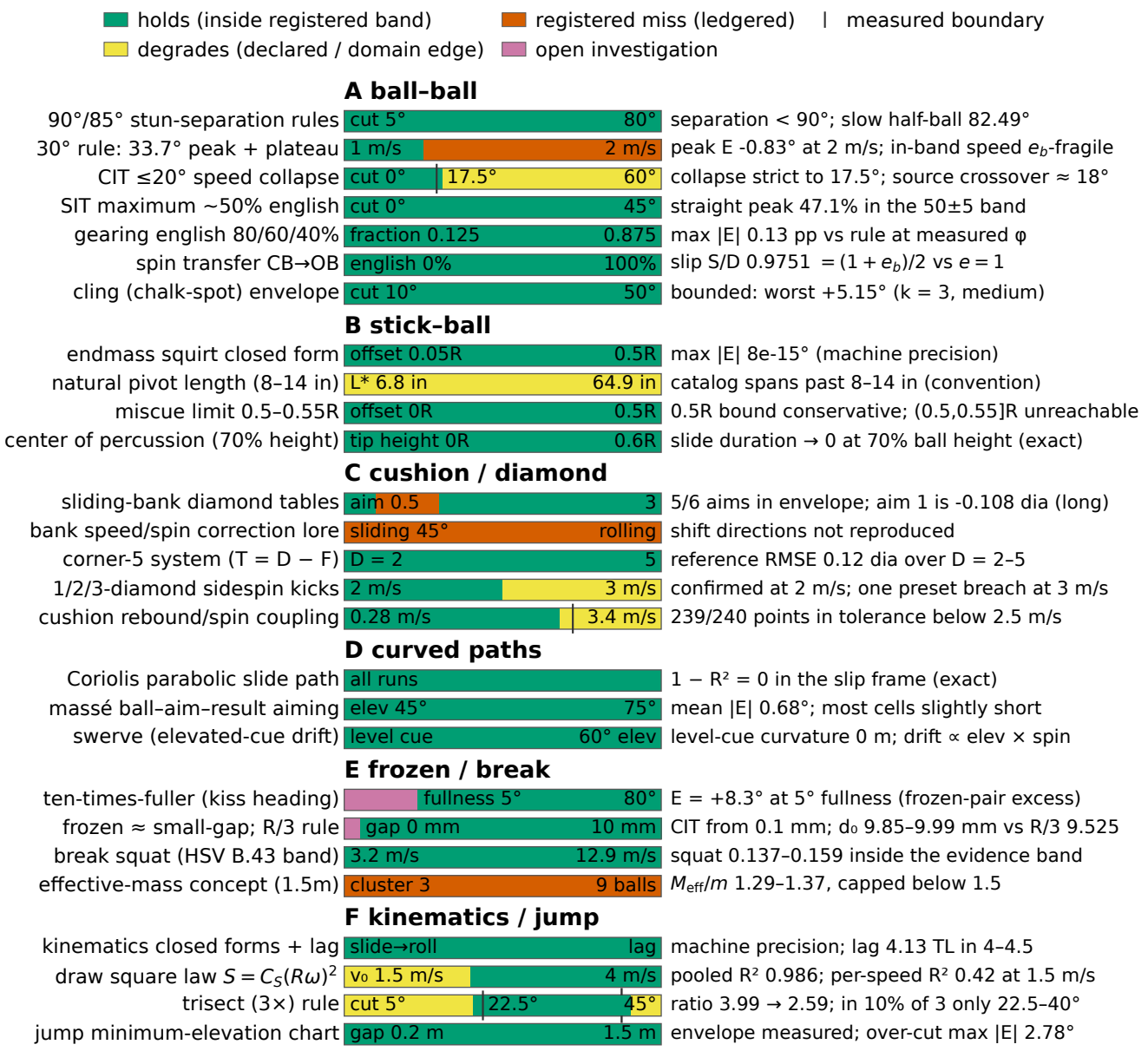
holds (inside registered band)
registered miss (ledgered)
measured boundary
degrades (declared / domain edge)
open investigation
A ball–ball
90°/85° stun-separation rules
cut 5°
80°
separation < 90°; slow half-ball 82.49°
30° rule: 33.7° peak + plateau
1 m/s
2 m/s
peak E -0.83° at 2 m/s; in-band speed e_b-fragile
CIT ≤20° speed collapse
cut 0°
17.5°
60°
collapse strict to 17.5°; source crossover ≈ 18°
SIT maximum ~50% english
cut 0°
45°
straight peak 47.1% in the 50±5 band
gearing english 80/60/40%
fraction 0.125
0.875
max |E| 0.13 pp vs rule at measured φ
spin transfer CB→OB
english 0%
100%
slip S/D 0.9751 = (1 + e_b)/2 vs $e = 1$
cling (chalk-spot) envelope
cut 10°
50°
bounded: worst +5.15° (k = 3, medium)
B stick–ball
endmass squirt closed form
offset 0.05R
0.5R
max |E| 8e-15° (machine precision)
natural pivot length (8–14 in)
L* 6.8 in
64.9 in
catalog spans past 8–14 in (convention)
miscue limit 0.5–0.55R
offset 0R
0.5R
0.5R bound conservative; (0.5,0.55]R unreachable
center of percussion (70% height)
tip height 0R
0.6R
slide duration → 0 at 70% ball height (exact)
C cushion / diamond
sliding-bank diamond tables
aim 0.5
3
5/6 aims in envelope; aim 1 is -0.108 dia (long)
bank speed/spin correction lore
sliding 45°
rolling
shift directions not reproduced
corner-5 system (T = D − F)
D = 2
5
reference RMSE 0.12 dia over D = 2–5
1/2/3-diamond sidespin kicks
2 m/s
3 m/s
confirmed at 2 m/s; one preset breach at 3 m/s
cushion rebound/spin coupling
0.28 m/s
3.4 m/s
239/240 points in tolerance below 2.5 m/s
D curved paths
Coriolis parabolic slide path
all runs
1 − R² = 0 in the slip frame (exact)
massé ball–aim–result aiming
elev 45°
75°
mean |E| 0.68°; most cells slightly short
swerve (elevated-cue drift)
level cue
60° elev
level-cue curvature 0 m; drift ∝ elev × spin
E frozen / break
ten-times-fuller (kiss heading)
fullness 5°
80°
E = +8.3° at 5° fullness (frozen-pair excess)
frozen ≈ small-gap; R/3 rule
gap 0 mm
10 mm
CIT from 0.1 mm; d_0 9.85–9.99 mm vs R/3 9.525
break squat (HSV B.43 band)
3.2 m/s
12.9 m/s
squat 0.137–0.159 inside the evidence band
effective-mass concept (1.5m)
cluster 3
9 balls
M_{eff}/m 1.29–1.37, capped below 1.5
F kinematics / jump
kinematics closed forms + lag
slide→roll
lag
machine precision; lag 4.13 TL in 4–4.5
draw square law $S = C_S(R\omega)^2$
v_0 1.5 m/s
4 m/s
pooled R² 0.986; per-speed R² 0.42 at 1.5 m/s
trisect (3×) rule
cut 5°
22.5°
45°
ratio 3.99 → 2.59; in 10% of 3 only 22.5–40°
jump minimum-elevation chart
gap 0.2 m
1.5 m
envelope measured; over-cut max |E| 2.78°